ELECTRONICS I PROJECT

Building a 2-to-1 Multiplexer From Scratch

Course:	Electronics I
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Abstract

This report documents the design and construction of a fully discrete 2-to-1 multiplexer using only NPN transistors and resistors. The circuit implements the Boolean function $Y = (D_0 \cdot \overline{S}) + (D_1 \cdot S)$ using Resistor-Transistor Logic (RTL). After initial debugging to resolve a grounding issue in the output stage, the circuit was successfully verified against its truth table. Finally, a potentiometer was used to sweep the Select input voltage, revealing the sharp threshold ($\approx 0.7\,\text{V}$) where the transistor physically transitions from OFF to ON.

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1 Introduction

In our digital logic course, we mostly looked at multiplexers as abstract black boxes. You have inputs, a select line, and an output, and the select line chooses which input passes through. They look clean on paper, but we wanted to see how they actually work using physical wires and transistors.

So, we decided to build one from scratch. The rule was simple: no integrated circuits allowed. Just discrete transistors and resistors. If the logic holds up, it should work regardless of what level you build it at.

This project was split into two parts. First, we built the MUX and made sure it worked. Second, we replaced the Select switch with a potentiometer to watch what happens physically when you slowly sweep the voltage across the transistor's threshold.

2 Theory and Logic Design

2.1 The Multiplexer Function

A 2-to-1 multiplexer routes one of two data inputs to a single output based on a select bit. The Boolean equation is:

$$Y = (D_0 \cdot \overline{S}) + (D_1 \cdot S) \tag{1}$$

When $S = 0$, the output equals D_0. When $S = 1$, the output equals D_1.

2.2 From AND-OR to NAND-NAND

Transistors are naturally good at making NAND gates, but not AND-OR structures directly. Using De Morgan's theorem, Equation (1) becomes:

$$Y = \overline{\overline{(D_0 \cdot \overline{S})} \cdot \overline{(D_1 \cdot S)}} \tag{2}$$

This requires:

- One inverter to generate $\overline{S}$
- Three NAND gates (two for the inner terms, one for the final combination)

Figure 1 shows the complete logic diagram.

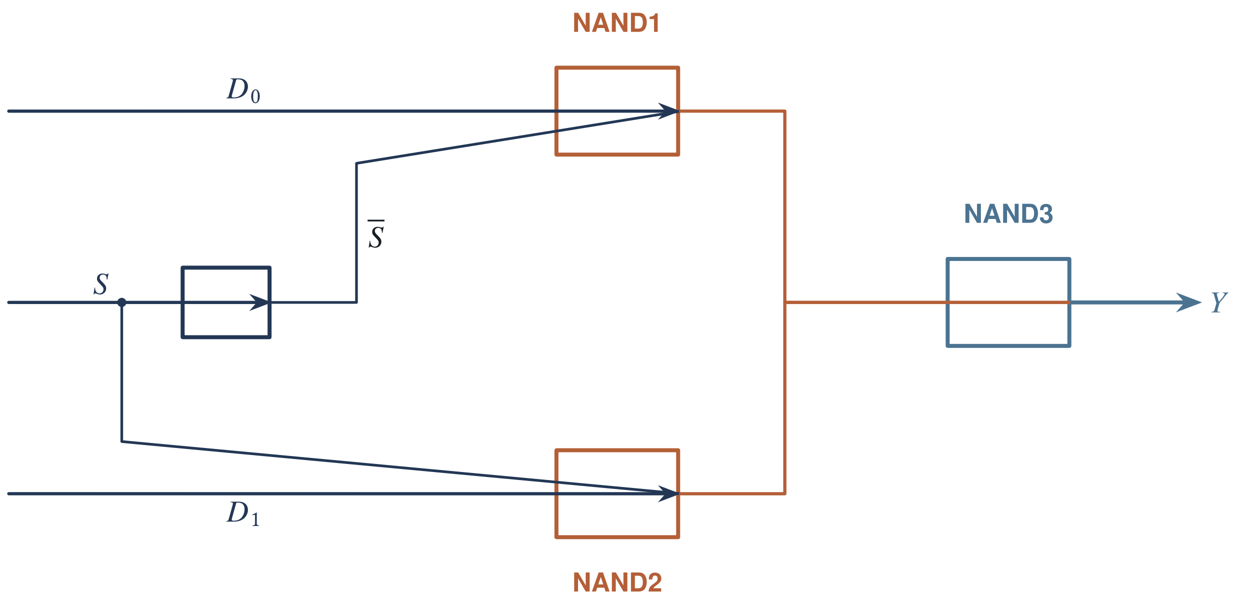

Figure 1 Logic diagram: NOT gate produces $\overline{S}$, NAND1 and NAND2 generate the intermediate terms, and NAND3 combines them for the final output.

2.3 Why NAND Works for Transistors

A NAND gate is built by placing two NPN transistors in series between the output node and ground. A pull-up resistor connects the output to +5V.

- If either transistor is OFF, the path to ground is broken. The output stays HIGH.
- Only when both transistors are ON does current flow to ground, pulling the output LOW.

This is perfect for RTL because a single transistor can block the entire path.

3 Transistor-Level Implementation

3.1 Stage 1: The Inverter (Q1)

The inverter uses one NPN transistor with a 1KΩ collector pull-up and a 10KΩ base resistor.

- Input LOW (0V): No base current. Transistor is OFF. The 1KΩ resistor pulls the output up to +5V.
- Input HIGH (5V): Base current flows. Transistor saturates. Output drops to roughly 0.2V.

3.2 Stage 2: NAND Gates (Q2-Q7)

Each NAND gate uses two transistors in series:

Table 1 NAND gate transistor mapping.

Gate	Transistors	Input A	Input B
NAND1	Q2 (top), Q3 (bottom)	D_0	$\overline{S}$
NAND2	Q4 (top), Q5 (bottom)	D_1	S
NAND3	Q6 (top), Q7 (bottom)	N_1	N_2

The top transistor's collector connects to the pull-up resistor. Its emitter connects to the bottom transistor's collector. The bottom transistor's emitter goes straight to ground.

3.3 Stage 3: Output Driver (Q8)

Q8 acts as a simple buffer. When Y is HIGH, base current flows through the 10KΩ resistor, Q8 turns ON, and the LED lights up. The 330Ω resistor limits the current to approximately 8mA to keep the LED safe.

3.4 Complete Circuit Schematic

Figure 2 shows the full transistor-level implementation. This is the diagram we used to wire the breadboard.

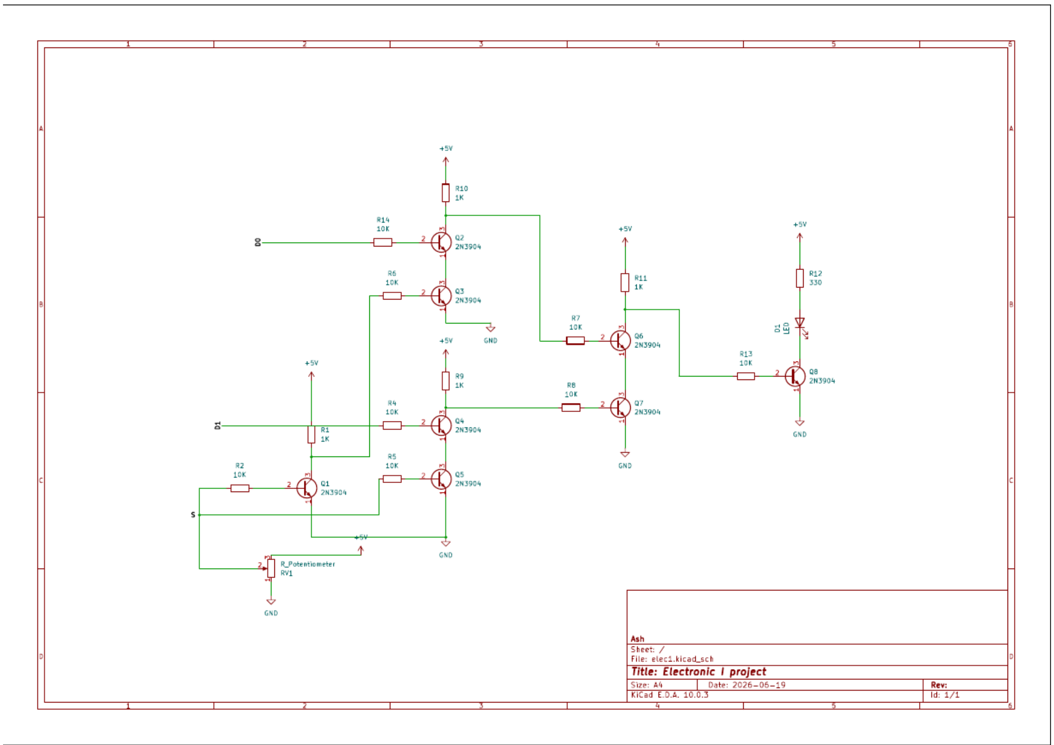

Figure 2 Complete RTL schematic of the 2-to-1 multiplexer. The circuit uses 8 transistors, 5 pull-up resistors (1KΩ), 8 base resistors (10KΩ), and one LED current-limiting resistor (330Ω).

4 Design Calculations

When building a circuit from discrete components, you cannot just guess resistor values. Every value affects how the transistor behaves. Here is how we chose ours.

4.1 Base Resistor (10KΩ)

The base resistor controls how much current enters the transistor base. For the transistor to act as a solid switch, it must be in *deep saturation*—not just barely on.

The collector current is set by the pull-up resistor:

$$I_C \approx \frac{V_{CC}}{R_C} = \frac{5\,\mathrm{V}}{1\,\mathrm{k\Omega}} = 5\,\mathrm{mA} \tag{3}$$

To guarantee saturation, we use a forced beta (β_{forced}) much lower than the transistor's actual current gain (typically 100–300 for 2N3904). We chose $\beta_{forced} \approx 12$:

$$I_B = \frac{I_C}{\beta_{forced}} = \frac{5\,\mathrm{mA}}{12} \approx 0.4\,\mathrm{mA} \tag{4}$$

Then the base resistor is:

$$R_B = \frac{V_{CC} - V_{BE}}{I_B} = \frac{5\,\mathrm{V} - 0.7\,\mathrm{V}}{0.4\,\mathrm{mA}} \approx 10.7\,\mathrm{k\Omega} \tag{5}$$

We used the standard value of 10KΩ.

4.2 Collector Pull-Up Resistor (1KΩ)

The pull-up resistor determines the logic HIGH level and the current drawn when the transistor is ON.

- **Logic HIGH:** When the transistor is OFF, the pull-up holds the output near 5V.
- **Power:** When the transistor is ON, current flows through the resistor to ground. With $R_C = 1\,\mathrm{k\Omega}$, this current is about 5mA—enough to drive the next stage but not so much that the transistor overheats.
- **Speed:** A smaller resistor charges the parasitic capacitance faster, but draws more power. 1KΩ is a standard compromise for RTL.

4.3 LED Current-Limiting Resistor (330Ω)

The LED needs a specific current to glow brightly without burning out. Red LEDs typically drop about 1.8V.

$$I_{LED} = \frac{V_{CC} - V_{LED} - V_{CE(sat)}}{R_{LED}} = \frac{5\,\mathrm{V} - 1.8\,\mathrm{V} - 0.2\,\mathrm{V}}{330\,\Omega} \approx 9\,\mathrm{mA} \tag{6}$$

This is a safe, bright current for a standard 5mm LED.

4.4 Power Supply (LM7805)

We used an LM7805 linear voltage regulator to provide a clean, stable +5V rail from a 9V battery. The LM7805 is a three-terminal regulator that maintains its output at exactly 5V regardless of load variations, as long as the input stays at least 2–3V above the output. This stability is important for RTL circuits because transistor thresholds and logic levels depend directly on the supply voltage.

5 The Breadboard Build

Building this on a breadboard was honestly the hardest part. The schematic is clean, but the physical breadboard turns into a mess of crossing wires really fast. We figured out a few things that made it easier to handle.

5.1 Wiring Strategy

We divided the breadboard into clean sections:

1. **Left side:** Input buttons and pull-down resistors
2. **Center-left:** Inverter Q1 and NAND1 (Q2, Q3)
3. **Center:** NAND2 (Q4, Q5)
4. **Center-right:** NAND3 (Q6, Q7)
5. **Right side:** LED driver Q8 and the output LED

5.2 The Breadboard Layout

Figure 3 shows our actual physical build.

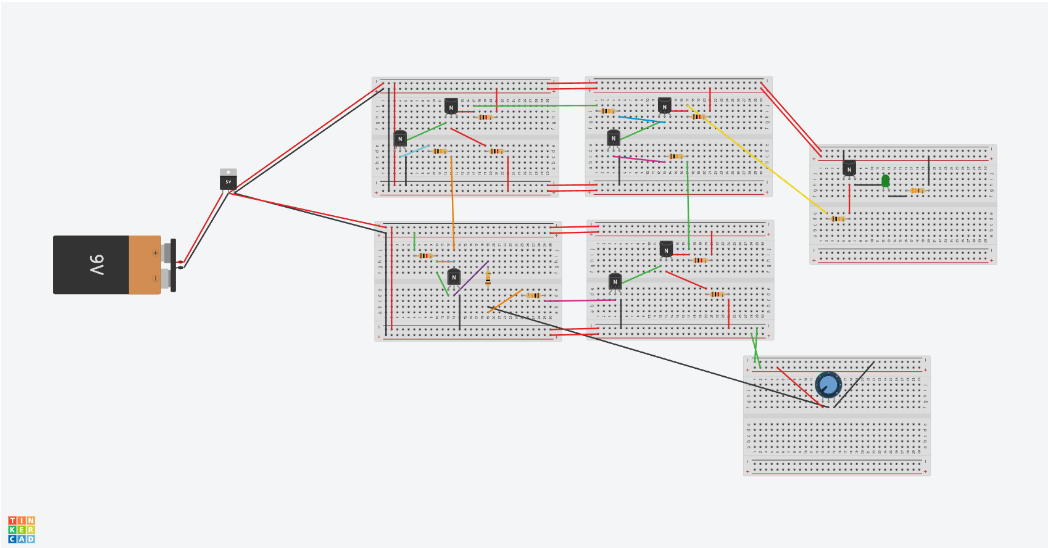

Figure 3 The MUX constructed on a solderless breadboard. Power is supplied by an LM7805 5V voltage regulator.

```
https://www.tinkercad.com/things/
4JDG91bPrdF-magnificent-leelo-luulia?
sharecode=P3w_
72XjgpzeE71dH-Q66N6LDBuPjUM-qISmru5L_
fw
```

Key Point

Setting D0 and D1 manually: In this prototype, the data inputs are not controlled by switches. Instead, you must wire D0 and D1 directly to either +5V (HIGH) or GND (LOW). Look at the four small breadboards arranged together. Ignore the board with the LED and the one with the potentiometer. The top-left board is for D0, and the bottom-right board is for D1. On each board, find the wire that connects the 10K base resistor to the transistor base. Move that wire to the +5V rail for a HIGH input, or to the GND rail for a LOW input.

5.3 Verification Measurements

After fixing the wiring, we verified every single node with a multimeter to be sure:

Table 2 Post-fix voltage measurements.

Node	Condition	Expected	Measured
Q1 Collector	$S = 5\text{V}$	$\approx 0\text{V}$	26 mV
Q2 Collector	$D_0 = 5\text{V}, S = 0\text{V}$	$\approx 5\text{V}$	4.61 V
Q4 Collector	$D_1 = 0\text{V}, S = 5\text{V}$	$\approx 5\text{V}$	4.61 V
Q6 Collector	$D_0 = 1, D_1 = 0, S = 0$	$\approx 0\text{V}$	57 mV

The small non-zero values (26mV, 57mV) are just the saturation voltage ($V_{CE(sat)}$) of the transistors. Because of the physical limits of the collector-emitter junction, it can't pull all the way down to a perfect, absolute 0V. This is totally normal.

6 The Threshold Experiment

6.1 The Setup

A potentiometer acts like an adjustable voltage divider. As you turn the knob, the wiper voltage changes smoothly from 0V up to 5V:

$$V_{wiper} = 5V \times \frac{R_2}{R_1 + R_2} \tag{7}$$

We connected this wiper to the S input (the base of Q1). Then we slowly turned the knob while watching the LED output.

6.2 What We Saw

At first, the LED showed the state of D_0. As we turned the knob up, nothing happened for a while. Then, sooner than we expected, the LED started to show D_1.

With the 250KΩ potentiometer, the transition happened slightly earlier than the typical 0.7V threshold. Because the high source resistance limits the base current, the transistor begins to conduct at a lower V_{BE} and enters saturation more gradually. The LED did not snap as sharply as it would with a lower-value potentiometer, but it still switched cleanly once the voltage crossed the active region.

Figure 4 shows this voltage transfer characteristic visually.

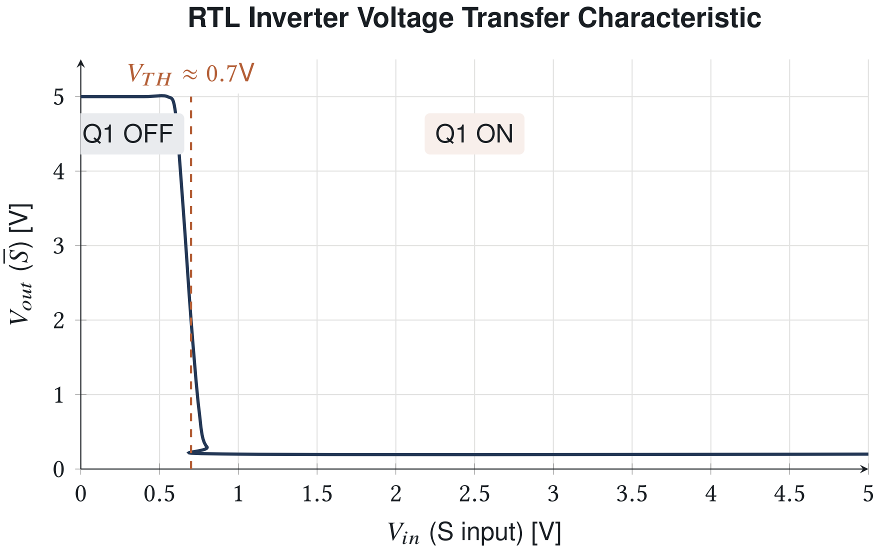

Figure 4 The output snaps from HIGH to LOW right around the base-emitter threshold. There's no slow fade because the transistor current rises exponentially with voltage.

Table 3 Behavior as V_S is swept from 0 to 5V.

V_S	Q1 State	$\overline{S}$	MUX Behavior
0V – 0.6V	OFF	HIGH (≈5V)	Selects D_0
≈0.7V	Active region	Transition	LED may flicker slightly
0.8V – 5V	ON (saturated)	LOW (≈0.2V)	Selects D_1

6.3 What This Means

The LED doesn't fade; it snaps. One moment it's passing D_0, the next moment it's passing D_1. That snap happens right at the transistor's physical threshold, not at some magical abstract logic boundary.

This is exactly what happens inside every modern digital chip. The "0" and "1" are just simple labels we put on voltages. The transistor doesn't actually know anything about digital logic—it only responds to voltage and current.

7 Results

We were able to verify the complete truth table with the working circuit:

Table 4 Verified MUX truth table.

S	D_0	D_1	Y	**LED**
0	0	0	0	OFF
0	1	0	1	ON
0	1	1	1	ON
1	0	0	0	OFF
1	0	1	1	ON
1	1	0	0	OFF
1	1	1	1	ON

When $S = 0$, the output perfectly follows D_0. When $S = 1$, the output perfectly follows D_1. This completely confirms the logic holds up in the physical build.

8 Bill of Materials

Table 5 Complete bill of materials for the discrete 2-to-1 MUX.

Ref	**Component**	**Value**	**Qty**	**Role**
Q1	NPN Transistor	2N3904	1	Inverter
Q2–Q3	NPN Transistor	2N3904	2	NAND1
Q4–Q5	NPN Transistor	2N3904	2	NAND2
Q6–Q7	NPN Transistor	2N3904	2	NAND3
Q8	NPN Transistor	2N3904	1	LED Driver
R1, R6, R9, R11	Resistor	1KΩ	4	Collector pull-up
R2, R4, R5, R7, R8, R10, R13	Resistor	10KΩ	7	Base resistor
R3	Resistor	1KΩ	1	Q1 collector pull-up
R12	Resistor	330Ω	1	LED current limit
LED1	LED	5mm red	1	Output indicator
RV1	Potentiometer	250KΩ	1	Select input sweep
(optional, I didn't use them) SW1–SW3	Push button	Tactile	3	D0, D1, S inputs
–	Breadboard	MB-102	1	Prototyping
–	Power supply	5V module	1	VCC rail

9 Conclusion

Building a 2-to-1 multiplexer entirely out of discrete transistors took more debugging than we originally expected, but that's where we learned the most. Tracking down the ungrounded Q7 emitter taught us how important it is to trace voltages node by node instead of just guessing where the problem is.

The threshold experiment ended up being the highlight of the project. By sweeping the Select voltage with a potentiometer, we got to see the exact physical moment the transistor crosses from OFF to ON, which is the foundational physical behavior that makes all digital circuits possible.

Project Repository

All project files, including the interactive circuit simulator, presentation slides, and build photos, are available on GitHub:

https://github.com/rowyfol/Elec101

Scan the QR code or visit the URL to access the full project repository.